

3. **Semi-Conductor Detectors:** A semi-conductor is a substance whose electrical conductivity is between that of a metal and an insulator. It is noted that Ge(Li) semi-conductors are excellent detectors of γ -rays with a resolution ten times higher than NaI (Th) scintillometers. The main disadvantage of these is a lower efficiency for higher energy x-rays. Besides, Ge(Li) semi-conductors need to be cooled by liquid nitrogen and are hence cumbersome and not suitable as field instruments.

Besides the above there are instruments known as γ -ray spectrometers, which can distinguish different energy peaks and hence make it possible to identify the source of radiation.

A Geiger counter will record "counts per minute", but this doesn't tell us what the radioactive substance is actually doing, merely what is reaching the detector. It also tells us nothing about the amount of damage being done to you. Thus we need several different units in order to measure radioactivity.

- The **activity** of a source is measured in **Becquerels (Bq)**,
One Becquerel is one decay per second.
- The amount of radiation that your **cells absorb** is measured in **grays (Gy)**,
One gray is one Joule of energy absorbed by 1kg of your body. This is the **dose** you receive.
- To measure the harm done to you, we need to remember that α particles ionise very strongly, and cause 20 times more cell damage than the same dose of β particles, γ rays or X-rays. We measure the "**dose equivalent**" in **sieverts (Sv)**.
- A dose of 1 gray of β -particles, γ -rays or X-rays will give you a dose equivalent of 1 sievert. A dose of 1 gray of α -particles will give you a dose equivalent of 20 sieverts

The sievert is quite a large unit, so we usually work in millisieverts (mSv) or microsieverts (μ Sv).

Here's a good way to think of it:

imagine you're out in a rainstorm.

**Then the amount of rain falling is measured in *Becquerels*,
the amount of rain hitting you is measured in *grays*,
and how wet you get is measured in *sieverts*.**

Our bodies have an activity of around 4000 Bq. The average annual dose in the UK is around 2.5 millisieverts.

Unit of Measuring Radioactivity:

The unit of measuring radioactivity is the Curie (Ci). One Curie is the amount of radiation produced by 1 gm of radium (Ra226). This is equivalent to 2.7×10^{11} decays per second. For practical measurements this unit is too large and therefore it has been replaced by the SI derived unit, the becquerel (Bq), which equates to one decay per second (therefore $1 \text{ Bq} = 2.7 \times 10^{-11} \text{ Ci}$ and 1 curie is 37 gigabecquerels).